

P2: EN retention after TL continuation (nanochat, WikiText-103 then TL-39) (#306935)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:

2026/08/17 07:58 (PT)

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

After EN0 (WikiText-103 nanochat) passes P0-E, does frozen P1.1 Tagalog continuation raise English held-out BPB more than matched extra English?

$C_{en} = \text{English val_bpb_full}(A2) - \text{English val_bpb_full}(A1)$. Predict $C_{en} \geq 0.01$ at d20. C_{en} is the declared Tagalog stream vs the declared extra-English stream (equal model-visible tokens), not pure language identity.

$G_{tl} = \text{Tagalog val_bpb_full}(A2) - \text{Tagalog val_bpb_full}(A1)$. Predict $G_{tl} \leq -0.01$. A0 to A2 Tagalog change is descriptive only.

A3 is a 50/50 document mix trade-off arm, not token-equated mitigation. Report $C_{en}(A3)$ and $G_{tl}(A3)$ with realized document/byte/token shares. Smaller C_{en} alone is not mitigation.

P0-E, before any Tagalog train token: EN0 English val_bpb_full is ≥ 0.01 below both an untrained same-depth model and a train-fitted add-one UTF-8 byte unigram on the same WT103 val bytes. If either fails, do not call the parent English-pretrained and do not interpret A2 as CF.

Do not claim P1.1 d20 forgot English. Do not load P1.1 weights as parent. Not CORE/chat/SFT. Does not amend #306780.

3) Describe the key dependent variable(s) specifying how they will be measured.

Primary DVs: English val_bpb_full on WT103-raw val; Tagalog val_bpb_full on frozen P1.1 reconstructed_article_70_15_15 val. Same English 32768 BPE. BPB = mean token NLL / (ln(2) * mean UTF-8 bytes per token), official evaluate_bpb (copy of P1.1 gate_j_full_bpb.py), T=2048, BOS-bestfit, full val (not --eval-tokens).

Contrasts at d20 val: C_{en} and G_{tl} as in Q2; A3 vs A1 the same way. Report unique documents, canonical bytes, unique BPE tokens, and revisit/epoch for A1/A2/A3 (not used to pick arms).

After d20 val is sealed, evaluate A2 d20 only once on official WT103-raw English test and once on frozen P1.1 test.jsonl (P2 English BPE). These are secondary treatment outcomes, not test-set C_{en}/G_{tl} (A1 is not tested), and do not select anything. P1.1 test.jsonl is a legacy external holdout, not a virgin P2 test. Do not reuse P1.1 test_bpb=1.164768.

4) How many and which conditions will participants be assigned to?

No human participants. Pin 92d63d4e8bb4df75c3b71618f31ddde2378b2bcd; tok_train -> base_train -> evaluate_bpb; T=2048; CORE off. No HF Trainer, llama.cpp, or python -m nanochat.dataset.

EN0: WT103-raw, English 32768 BPE, d8 and d20, $D_{3x_en} = 3 * T_{en_train}$, $N_{EN0} = \text{ceil}(D_{3x_en}/65536)$, final ckpt.

A0: freeze EN0 d20. A1/A2/A3 required at d20 only from EN0 d20. d8 continuation is exploratory.

A1: extra English, official WT103-raw train. A2: frozen P1.1 Tagalog train. A3: 50/50 documents, not tokens. Freeze mix before any EN0 val_bpb_full: seed 42, sha256-sort each language, take $K = \min(n_{en}, n_{tl})$, interleave EN/TL, cycle, pack to D_{phase2} . Report realized shares.

$D_{phase2} = 294 * 65536 = 19267584$ English-BPE tokens. Fresh Adam. LR peak = $0.3 * EN0$ peak. Warmup < N (default 14 if $14 < N$). Never load P1.1 model_000294.pt.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

P0-E at d8 and d20 before any Tagalog train token (untrained and byte-unigram floors, 0.01). If P0-E fails, stop the CF claim.

Primary contrasts at d20 final val only: C_{en} and G_{tl} . A0 Tagalog val is descriptive. A3: report $C_{en}(A3)$, $G_{tl}(A3)$, mix shares; do not call smaller C_{en} mitigation.

One seed: point estimates only. No seed interval, token-level CI, or significance test. $|C_{en}|$ or $|G_{tl}| < 0.01$ is not a ranking. 0.01 is a practical cutoff. No CORE in the confirmatory comparison. A2 English trajectory is descriptive. 1/5/10% replay, FilBench, EWC, tokenizer swap, CORE/SFT are exploratory.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Drop a unit only if null/empty after LF, or length > 200000 chars; if drops > 5%, stop. Official WT103-raw splits. Frozen P1.1 Tagalog train SHA256

2b0474c5700dc1eba14def572aa23cc227e4c59c10c2de3ce6b7bda75d137687. Split invalid if any exact hash is in more than one of train/val/test per language.

Exclude a run only for: NaN/Inf; ClimbMix; val/test in train dir; new tokenizer after BPB; ratio -1; HF Trainer; writing into p1-fixed-d20-3x; loading P1.1 model_000294.pt; P0-E-fail called English-pretrained; P1.1 test_bpb reused as A2; attention patch; warmup $\geq$ iterations.

Do not drop an arm because C_en or G_tl looks bad. Do not relabel A3 as mitigation after seeing shares. d4 smoke is not confirmatory.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

English: Salesforce/wikitext, wikitext-103-raw-v1, official splits, no added docs. Tagalog: frozen P1.1 reconstructed_article_70_15_15 train, no rebuild.

EN0: one seed, d8 and d20, $N_{EN0} = \text{ceil}(3 \cdot T_{en_train} / 65536)$, $B=65536$, $T=2048$.

A1/A2/A3: one seed each at d20, $N=294$, $D_{\text{phase2}}=19267584$. d8 continuation exploratory.

After d20 val seal: test A2 d20 only, one English WT103-raw test_bpb and one Tagalog P1.1 test.jsonl BPB (P2 English BPE), P2 test-access log. A1 is not test-evaluated.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Q1: P1.1 corpus and P1.1 outcomes exist. No P2 EN0/A1/A2/A3 or P2 English-BPE evaluation outcome has been computed.

New study. Does not amend #306780. Reuses frozen P1.1 Tagalog files as documented inputs; does not reuse P1.1 confirmatory BPB as P2 observations.

Descriptive: unique docs/bytes/tokens/epochs; A2 English trajectory; PTPP R_d; fertility; P1.1 d20 on English UTF-8 as never-trained-on-English OOD (not BWT).

Exploratory: d8 A1/A2/A3; 1/5/10% replay; FilBench; EWC; tokenizer swap; CORE; SFT; Spark/MPS.

Protocol SHA256 da7ac07b86d097770c10e76c79f0534d6c609ceea71baabcf587a4fcd01bc0e8. Pin 92d63d4. Test files stay outside train dirs.